SF2940 Probability: Exam

Fall 2023

Date: 23. October 2023, 08:00-12:00.
Examiner: Kevin Schnelli, SCI MAT. Phone: 72 02
Aids: One page hand written notes. No other aids are allowed. Formulary with distributions is included after the problem statement.

General instructions: Your solutions and answers should be legible, easy to follow and not lack in explanations. For full points a correct and well-presented reasoning is required.

Exam design: This exam consists of four independent problems. Each problem gives at most 5 points.

Other: Solutions will be posted on Canvas within a week after the exam.

Problem 1. Let $X_1, X_2, \dots$ be a sequence of random variables, let X be another random variable.

a.) Define what it means that X_n converges in square mean to X, as $n \to \infty$. ..... (1point)

b.) Assume that X_n converges in square mean to X, as $n \to \infty$. Show that X_n converges in probability to X, as $n \to \infty$. ..... (1point)

c.) Assume further that $X \in Exp(1)$. Explicitly compute the moment generating function $\psi_X(t)$, and determine $h > 0$ such that for all t satisfying $|t| < h$, the moment generating function exists. ..... (2points)

d.) For all $k = 1, 2, 3, \dots$, compute the k-th moment of X, i.e., $\mathbb{E}[X^k]$. ..... (1point)

Problem 2. Let the random variables X and Y have joint probability density

$$f_{X,Y}(x,y) = \begin{cases} 4\mathrm{e}^{-2y}\,, & 0 < x < y\,, \\ 0\,, & \text{else}\,. \end{cases}$$

a.) Determine the conditional expectation $\mathbb{E}[X|Y]$. . . . . (2points)

Next, set

$$U := Y\,, \qquad V := \frac{X}{Y - X}\,.$$

b.) Find the joint probability density $f_{U,V}(u,v)$ of U and V. . . . . (3points)

Problem 3. Let $\mathbf{X} = (X_1, X_2, X_3)'$ be a three-dimensional Gaussian vector with mean vector identically equal to zero and covariance matrix

$$\Lambda_{\mathbf{X}} = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 3 & -1 \\ 1 & -1 & 2 \end{pmatrix}\,.$$

Set now $Y_1 := X_1 + X_2 + X_3$ and $Y_2 := X_1 - X_3$.

a.) Determine the distribution of the random vector $\mathbf{Y} = (Y_1, Y_2)'$. . . . . (2points)

b.) Determine the conditional density function of Y_1 given that $Y_2 = 0$.
. . . . . (2points)

c.) Compute the expectation of the random variable $M := \max\{X_1, X_3\}$, i.e., find $\mathbb{E}[M] = \mathbb{E}[\max\{X_1, X_3\}]$. *Hint:* The random variables X_1 and X_3 are not independent. But we can use that for any real numbers $a, b \in \mathbb{R}$, we have that

$$\max\{a,b\} = \frac{1}{2}(a+b) + \frac{1}{2}|a-b|\,.$$

. . . . . (1point)

Problem 4. Let $X_1, X_2, \ldots,$ be a sequence of i.i.d. random variables with probability density function

$$f_{X_1}(x) = \begin{cases} 2(1-x)\,, & \text{if } 0 < x < 1\,, \\ 0\,, & \text{else}\,. \end{cases}$$

Introduce now

$$M_n := \min\{X_1, X_2, \ldots, X_n\}\,, \qquad n = 1, 2, 3, \ldots\,.$$

a.) Show that M_n converges in probability to zero, as $n \to \infty$.
Hint: Carefully show first that

$$\mathbb{P}(M_n > y) = \big(\mathbb{P}(X_1 > y)\big)^n\,.$$

..... (2points)

b.) Show that the sequence

$$Y_n := nM_n\,,$$

converges in distribution as $n \to \infty$, and determine the limiting distribution.
..... (2points)

c.) Let N be a discrete random variable taking values in the positive integers $n = 1, 2, 3, \ldots$, that is independent from $X_1, X_2, \ldots$. Set now

$$M_N := \min\{X_1, X_2, \ldots, X_N\}\,.$$

Show that the cumulative distribution function of M_N satisfies

$$\mathbb{P}(M_N \leq y) = 1 - g_N\big(\mathbb{P}(X_1 > y)\big)\,, \qquad y \in \mathbb{R}\,,$$

where g_N is the probability generating function of N..... (1point)

Good luck!

Distribution	Notation	Probability function	$\varphi_X(t)$
One point	$\delta(a)$, $a \in \mathbb{R}$	$p(a) = 1$	$\mathrm{e}^{\mathrm{i}ta}$
Bernoulli	$\mathrm{Be}(p)$; $0 \le p \le 1$	$p(0) = q$, $p(1) = p$; $q = 1 - p$	$q + p\mathrm{e}^{\mathrm{i}t}$
Binomial	$\mathrm{Bin}(n, p)$, $n = 1, 2, \ldots$; $0 \le p \le 1$	$p(k) = \binom{n}{k} p^k q^{n-k}$, $k = 0, 1, 2, \ldots, n$; $q = 1 - p$	$(q + p\mathrm{e}^{\mathrm{i}t})^n$
Geometric	$\mathrm{Ge}(p)$, $0 \le p \le 1$	$p(k) = pq^k$, $k = 0, 1, 2, \ldots$; $q = 1 - p$	$\frac{p}{1-q\mathrm{e}^{\mathrm{i}t}}$
Poisson	$\mathrm{Po}(\lambda)$, $\lambda > 0$	$p(k) = \mathrm{e}^{-\lambda}\frac{\lambda^k}{k!}$, $k = 0, 1, 2, \ldots$	$\mathrm{e}^{\lambda(\mathrm{e}^{\mathrm{i}t}-1)}$

Table 1. Discrete distributions

Distribution	Notation	Density	$\varphi_X(t)$
Uniform	$\mathrm{U}(a, b)$, $a < b$	$f(x) = \frac{1}{b-a}$, $a < x < b$	$\frac{\mathrm{e}^{\mathrm{i}tb}-\mathrm{e}^{\mathrm{i}ta}}{\mathrm{i}t(b-a)}$
Exponential	$\mathrm{Exp}(\lambda)$, $\lambda > 0$	$f(x) = \frac{1}{\lambda}\mathrm{e}^{-x/\lambda}$, $x > 0$	$\frac{1}{1-\lambda\mathrm{i}t}$
Laplace	$\mathrm{L}(\lambda)$, $\lambda > 0$	$f(x) = \frac{1}{2\lambda}\mathrm{e}^{-\lvert x\rvert/\lambda}$, $x \in \mathbb{R}$	$\frac{1}{1+\lambda^2t^2}$
Beta	$\beta(r, s)$, $r, s > 0$	$f(x) = \frac{\Gamma(r+s)}{\Gamma(r)\Gamma(s)}x^{r-1}(1-x)^{s-1}$, $0 < x < 1$	$*$
Normal	$\mathrm{N}(\mu, \sigma^2)$, $\mu \in \mathbb{R}$, $\sigma > 0$	$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}}\mathrm{e}^{-\frac{1}{2}(x-\mu)^2/\sigma^2}$, $x \in \mathbb{R}$	$\mathrm{e}^{\mathrm{i}\mu t-\frac{1}{2}\sigma^2t^2}$
Standard normal	$\mathrm{N}(0, 1)$	$f(x) = \frac{1}{\sqrt{2\pi}}\mathrm{e}^{-\frac{1}{2}x^2}$, $x \in \mathbb{R}$	$\mathrm{e}^{-\frac{1}{2}t^2}$
Log-normal	$\mathrm{LN}(\mu, \sigma^2)$, $\mu \in \mathbb{R}$, $\sigma > 0$	$f(x) = \frac{1}{x\sqrt{2\pi\sigma^2}}\mathrm{e}^{-\frac{1}{2}(\log x-\mu)^2/\sigma^2}$, $x > 0$	$*$
Cauchy	$\mathrm{C}(m, \eta)$, $\eta > 0$, $m \in \mathbb{R}$	$f(x) = \frac{1}{\pi}\frac{\eta}{(x-m)^2+\eta^2}$, $x \in \mathbb{R}$	$\mathrm{e}^{\mathrm{i}mt-\eta\lvert t\rvert}$

Table 2. Continuous distributions